

COMMON QUESTIONS

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
7/1	(a)	(i)		sulfuric acid accept H ₂ SO ₄		1		1		1
		(ii)	I	fizzing / bubbling / effervescence neutral answer – temperature change / colour change	1			1		1
			II	award (1) for any indication of the acid being used up e.g. to use up (all) the (sulfuric) acid to neutralise (all) the (sulfuric) acid so that no acid is left do not accept – to use up all the reactants	1			1		1
		(iii)		filter (the mixture) / filtration (1) award (1) for any indication of evaporation or steps taken for evaporation to happen e.g. evaporate (solution) heat / boil (solution) leave (solution) in a warm place / on window sill leave (solution) for a day / until next lesson neutral answer – leave / leave to form crystals	2			2		2
		(iv)		ZnSO ₄		1		1		

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					AO1	AO2	AO3	Total	Maths	Prac																		
	(b)	(i)		<table><caption>Data points from the graph</caption><tr><th>x</th><th>y</th></tr><tr><td>0</td><td>20.5</td></tr><tr><td>5</td><td>22.0</td></tr><tr><td>10</td><td>23.5</td></tr><tr><td>20</td><td>26.5</td></tr><tr><td>25</td><td>26.3</td></tr><tr><td>30</td><td>25.9</td></tr><tr><td>35</td><td>25.5</td></tr><tr><td>40</td><td>25.2</td></tr></table> award (2) for all four points plotted correctly – tolerance $\pm\frac{1}{2}$ square award (1) for any two correct points award (1) for straight line	x	y	0	20.5	5	22.0	10	23.5	20	26.5	25	26.3	30	25.9	35	25.5	40	25.2		3		3	3	3
x	y																											
0	20.5																											
5	22.0																											
10	23.5																											
20	26.5																											
25	26.3																											
30	25.9																											
35	25.5																											
40	25.2																											
		(ii)	I	24.9 accept correct value from graph		1		1	1																			
			II	accept any value in the range 20.5 – 21.0 accept correct value from graph			1	1	1																			

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		(iii)		award (1) for each correct answer use a beaker instead of a flask ☐ repeat the method ☐ add the acid in smaller intervals ☐ wrap cotton wool around the flask ☒ use a larger flask ☐ place a lid on the flask ☒			2	2		2
	(c)			award (2) for either of following sodium hydroxide is twice as concentrated (as hydrochloric acid) hydrochloric acid is half as concentrated (as sodium hydroxide) award (1) for either of following sodium hydroxide is more concentrated (than hydrochloric acid) hydrochloric acid is less concentrated (than sodium hydroxide) accept reference to strength instead of concentration			2	2		2
				Question 7/1 total	4	6	5	15	5	12